1. Assumptions for Q1–Q15

Q1. Basic filtering and sorting

- Only orders with `Order_Status = 'Delivered'` are included.
- The date range is inclusive, from 1 October 2024 through 31 March 2025.
- Each row represents one order.
- Orders are sorted by `Order_Date` in descending order, with the latest first.

Q2. Multi-table JOIN

- Only delivered orders are considered.
- Each order must have a matching customer record and employee record.
- `INNER JOIN` excludes orders with missing customer or employee matches.
- Each order is associated with at most one customer and one employee.

Q3. Aggregation and HAVING

- Only delivered orders contribute to units sold and net revenue.
- Net revenue is calculated after discounts:

Net Revenue = Quantity * Sales Price * (1 − Discount Rate).

- `Discount_Rate` is stored as a decimal; for example, `0.10` means a 10% discount.
- Categories are included only when their total net revenue is strictly greater than ₹10,000.
- Units sold means the sum of item quantities, not the number of orders.

Q4. Customers with repeat purchases

- A repeat customer has at least two distinct delivered orders.
- Returned and cancelled orders are excluded based on their current `Order_Status`.
- Total net spend includes discounts.
- Customers without delivered purchases are excluded.

Q5. LEFT JOIN and zero activity

- All customers must appear, even those with no delivered orders.
- Only delivered orders are counted.
- Customers without delivered orders receive a count of zero.
- The delivered-status condition must remain in the `ON` clause of the `LEFT JOIN` to preserve customers with zero activity.

Q6. Subquery

- The average unit price is calculated across all products in the `Products` table.
- Each product's `Unit_Price` is compared with that overall average.
- Only products with a unit price strictly greater than the average are returned.
- Every product is considered once, regardless of sales or stock quantity.
- The average is calculated using the current product prices, not historical sale prices.

Q7. NOT EXISTS

- A customer qualifies if they have never placed an order currently marked as `Delivered`.
- Customers with no orders are included.
- Customers whose orders are exclusively cancelled or returned are included.
- The query checks the current order status, not the historical delivery status.

Q8. Date functions

- Only delivered orders contribute to the monthly order count and net revenue.
- Orders are grouped by calendar year and month, so September 2024 and September 2025 are treated as different months.
- Net revenue is calculated after discounts.
- The order count counts distinct orders, not order items.
- Only months containing at least one delivered order appear.

Q9. CASE expression

- Stock quantities are classified according to the specified thresholds:
- Below 20: `Low Stock`
- 20 through 40 inclusive: `Moderate Stock`
- Above 40: `High Stock`
- The classification uses current `Stock_Quantity`.
- Stock quantity is assumed to be a valid, non-negative integer.
- No stock quantity is `NULL` in the supplied sample data.

Q10. Conditional aggregation

- Every employee is included, even employees without orders.
- Orders are counted according to their current status: `Delivered`, `Returned`, or `Cancelled`.
- Each order is counted once, regardless of how many items it contains.
- Employees with no matching orders receive zero in all three counts.
- Any other order statuses would not contribute to these three counts.

Q11. Window function — ranking

- Every product is eligible to appear, including products with no delivered sales.
- Only delivered orders contribute to net revenue.
- Products with no delivered sales are assigned zero revenue.
- Products are ranked separately within each category.
- `RANK()` gives tied products the same rank, with gaps in subsequent rank numbers.
- Revenue is calculated after discounts.
- Products with a `NULL` category would form a separate category partition.

Q12. Running total

- Only delivered orders contribute to monthly revenue.
- Monthly net revenue is calculated after discounts.
- Months are ordered chronologically using the calendar month.
- The running total starts with the earliest month in the result and accumulates revenue through each subsequent month.
- Months without delivered orders are omitted unless a separate calendar table or generated month series is used.
- The running total is cumulative across all months in the dataset, not reset each year.

Q13. Return analysis

- The `Returns` table identifies returned orders and provides their return dates and reasons.
- Pre-discount order value is calculated as quantity multiplied by sale price, without subtracting discounts.
- Each returned order is assumed to have at least one order item.
- The query reports the full pre-discount value of the order, not necessarily the value of the specific item returned.
- Refund amounts are unavailable because the `Returns` table does not contain a refund amount.
- The query does not independently filter `Order_Status = 'Returned'`; it relies on records in the `Returns` table.

Q14. CTE — customers above average revenue

- The CTE calculates net revenue for each customer using delivered orders only.
- Net revenue includes the effect of discounts.
- The average is calculated **only across customers with at least one delivered purchase**, not across all customers.
- Customers with no delivered purchases are excluded from both the average and the final result.
- Only customers whose revenue is strictly greater than the average are returned.
- The average is an unweighted arithmetic mean of customer revenue totals.

Q15. Business insight — top three customers

- Only delivered orders contribute to customer net spend.
- Net spend is calculated after discounts.
- Delivered order count counts distinct orders rather than order items.
- Revenue contribution is calculated as:

$$\text{Customer Net Spend}\text{Total Delivered Net Revenue}\times 100\text{Total Delivered Net Revenue}\text{Customer Net Spend}\times 100$$

- The denominator includes delivered net revenue from all customers, not just the top three.
- Customers are ranked by net spend in descending order.
- `LIMIT 3` returns at most three customers. If customers tie at the third position, the query does not necessarily return all tied customers.
- Returned and cancelled orders are excluded according to their current status.

Important assumptions that apply to all questions

These are useful to mention at the beginning of your assignment:
Order status: `Order_Status` represents the current status of an order.

Discounts: `Discount_Rate` is stored as a decimal between 0 and 1.

Revenue: Net revenue means revenue after discounts, before any separately recorded refunds, taxes, shipping charges, or other adjustments.

Historical prices: `Unit_Price_at_Sale` is used for revenue calculations because it represents the price at the time of sale; `Products.Unit_Price` is used for the average-price question.

Data integrity: Primary keys and foreign keys are assumed to be valid, and each order item belongs to one order and one product.

Returns: Returned orders are excluded from delivered-order calculations based on their current status. No refund is deducted unless refund data is available.

Currency: All monetary amounts are assumed to be in Indian rupees (₹), as indicated by the assignment.

Exam tip: If you need to write only a few assumptions, prioritize the definition of delivered orders, the revenue formula, discount representation, treatment of zero-activity customers, and the population used to calculate averages.